Taller 3 circuito 2
317

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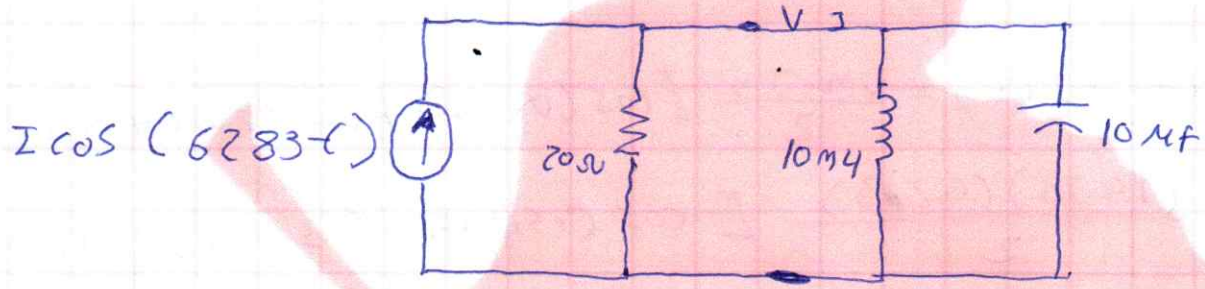
1) valor de Potencias istananeas del inductor

• $\omega = 6283$ [rad/s] • $I_s = 1 \cdot \cos(\omega t)$ A




$Z_R = 20\Omega$

$Z_L = j\omega L = j6283\,(0{,}01)$

$Z_L = 62.83j\,\Omega$

$Z_C = \frac{1}{j\omega C} = \frac{1}{j(6283)(10\times10^{-6})} = -j15.91\,\Omega$

$I = 1\angle 0$

se Hallo corriente en Zadudor

$I_L = \frac{14.70\angle -43.19}{j62.53} = 0.23\angle -133.19$ [A]

$\frac{V_1}{20} + \frac{V_1}{62.83j} + \frac{V_1}{-j15.91} = 1\angle 0$

$V_1\, 0.068\angle 43.19 = 1\angle 0$

$V_1 = \frac{1\angle 0}{0.068\angle 43.19}$

$V_1 = 14.70\angle -43.19$ [V]

mismo voltaje en las ramas

$D(t) = V(t) \cdot I(t)$ , $V1(t) = 14.70 \cos(6283t - 43.19)$

$I2(t) = 0.23 \cos(6283t - 133.19)$

$P(t) = 14.70 \cos(6283t - 43.19) \cdot (0.23 \cos(6283t - 133.19)]$

Formula

$P(t) = V(t) \cdot I(t) = \frac{Vm \cdot Im}{2} [\cos(\alpha - \beta) + \cos(2\omega t + \alpha t \beta)]$

$P(t) = \frac{(14.70)(0.23)}{2} [\cos(-43.19 - (-133.19)) + \cos(2\omega t - 43$

$- 133.19)]$

$P(t) = 1.69 [\cancel{\cos(90^\circ)} + \cos(2\omega t - 176.38]$

$P(t) = 1.69 \cos(2\omega t - 176.38)\ W$

$\downarrow$

$P = I$

potencia inductancia

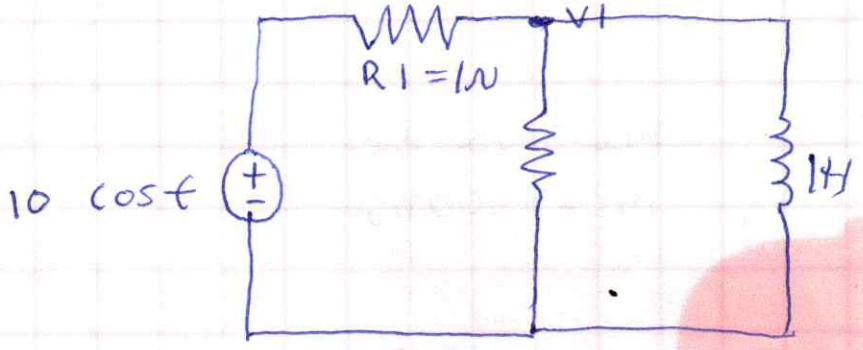
Potencia Activa o promedio

$P = \frac{1}{2} (14.70)(0.25) \cos[-43.19 - (-133.19)]$

$P = 1.69 \cancel{\cos(90^\circ)}$

$P = 0.W$

2 Potencia Actuta ó media)




• 10 cos (t) entonces $\omega = 1$

$\omega = 1 Rad/s$

$$V1 - \frac{10\angle 0}{1} + \frac{V1}{1} + \frac{V1}{j1} = 0$$

$$\frac{V1}{1} + \frac{V1}{1} + \frac{V1}{j1} = 10\angle 0$$

$$2.23\angle -26.56\, V1 = 10\angle 0$$

$$V1 = 4.48\angle 26.56\ [V]$$

Inductancia

$$ZL = j\omega L = j(1)(1) = j1\Omega$$

Fuente $= 10\angle 0$

corriente en R1

$$IR1 = \frac{6.31\angle -18.48}{1} =$$

$$IR1 = 6.31\angle -18.48\ [A]$$

voltaje en R1

$$VR1 = 10\angle 0 - 4.48\angle 26.56$$

$$VR1 = 6.31\angle -18.48\ [V]$$

$$P = \frac{1}{2}(6.31)(6.31)\cos[-18.48-(18.48)]$$

$$P = 19.90\cos(0) = 19.90\,W \approx 20\,W$$

+R1 potencia

$$Zp1 = \frac{1 \cdot j1}{1+j1} = 0.70\angle 45$$ → Resultado de potencia paralela

Actuta R1 + Zp1

$$Zp_0 = 1 + 0.70\angle 45 = 1.57\angle 18.31$$

Hallamos la corriente total

$$I1 = \frac{10\angle 0}{1.57\angle 18.31} = 6.36\angle -18.31$$

entonces la potencia es

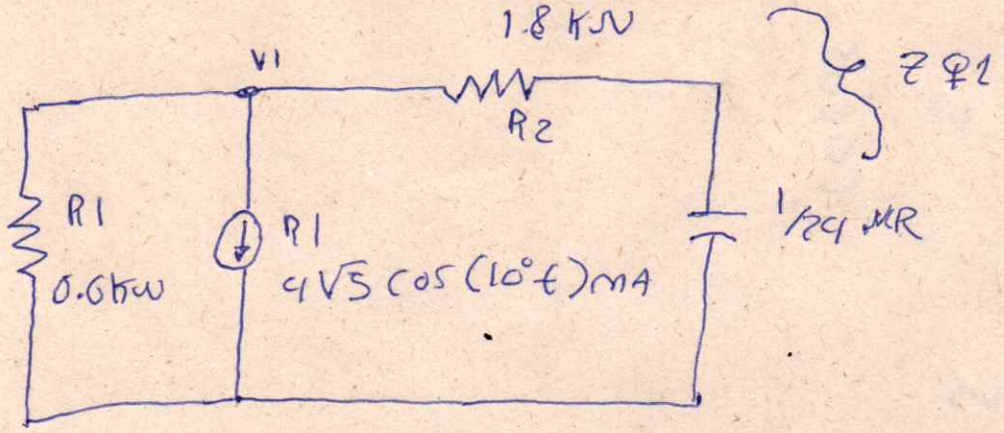
$$P = \frac{1}{2}(10)(6.36)\cos[0-(-18.31)]$$

$$P = 31.8\cos(18.31)$$

$$P = 30.18\,W \approx 30\,W$$

GOD GIVE HIS REAVEN ALL'S RIGHT WITH THE WORLD

b Determinar potencia




$R1 = 60$ $\omega = 10$

$R2 = 1800\Omega$

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j(10.000)(\frac{1}{24}\mu f)} = j\,2,4390.2\,\Omega$$

$Z_{q1} = 4390236\angle 90\,\Omega$

$f1 = 0.00894\angle 0$

$$\frac{V1}{600} + \frac{V1}{4390236\angle 90} = 0.00894\angle 0$$

$V1\,0.001666\angle -0.007 = -0.00894\angle 0$

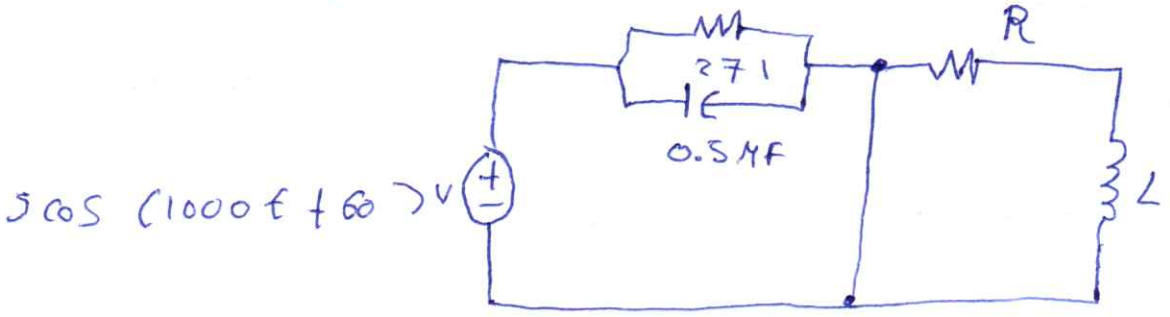
$V1 = 5.36\angle -179.99$

Abero
corriente IR1

$$IR1 = \frac{5.36\angle -179.99}{600}$$

$= 0.008\angle -179.99$ [A]

3 transferencia maxima de Potencia




$$Z_{th} = \frac{4000 - j2000}{4000 - j2000} = \frac{-j8 \times 10^6}{4000 - j2000}$$

$F_1 = 5 \angle 6$ V

$\omega = 1000$ rad/s

$$Z_C = \frac{1}{j(1000)(0.5 \times 10^{-6})}$$

$Z_C = -j2000\,\Omega$

$4000 + j2000$ Para obtener $a + jb$

real imaginario

Entonces

$$Z_{th} = \frac{-j8 \times 10^6}{4000 - j2000} - \frac{4000 + j2000}{4000 + j2000} = \frac{-j32 \times 10^9 + 16 \times 10^9}{16 \times 10^6 + j8000 - j8000 + 4000}$$

$$= \frac{-j32 \times 10^9}{20 \times 10^6} + \frac{16 \times 10^9}{20 \times 10^6}$$

Potencia Activa en R1

$$P = \frac{1}{2}(5.36)(0.008)\cancel{\cos}(-179.99-(-179.99))$$

$$P = 0.0244\,W \rightarrow [24.4\,mW]$$

Ahora en f1

$$P = \frac{1}{2}(5.36)(0.00844)\cos[-179.99-(0)]$$

$P = -0.023\,W \rightarrow$ potencia consumida

Ahora en :

$$I_{zq} = \frac{5.36\angle 179.99}{4390236\angle 90} = 0.00000122\angle 89.99$$

$$V_C = \frac{0.00.00122\angle 89.99}{-j2439.02} = 0.00297\angle 179.99$$

$$P = \frac{1}{2}(0.00297)(0.0000122)\cos[\cancel{89.99-179.99}]$$

$$P = 0\,W$$

$$Z_{th} = \frac{-j32\times10^4}{20\times10^6} + \frac{16\times10^4}{20\times10^6} = 800 - j1600\,\Omega$$

$$Z_{eq} = R + j\omega L$$

entonces

$R = 800 -$ parte real $= R$

$Z_L = -j1600 -$ parte imaginaria

Para hallar $L$: $L = \frac{1600}{\omega} = \frac{16\cancel{00}}{10\cancel{00}} = 1.6\,H$

Datos

$F1 = 5\angle 45$

$\omega = 500$ rad/s

$Z_L = j(5000)(0.35)$

$= j1600$

equivalencia de parte en serie

$Z_{eq1} = 800 + j1600$

Impedancia del capacitor

$Z_C = -jX_C \quad X_C = \frac{1}{j\omega C}$

Resistencia capacitativa

$$Z_{eq2} = \frac{R - Z_C}{R + Z_C} =$$

$$Z_{eq2} = \left(\frac{1}{4000} + j500C\right)^{-1} = \frac{1}{800 - j1600}$$

$$\frac{1}{800 - j1600} = \frac{800 + j1600}{800 + j1600} = \frac{800 + j1600}{640.000 \cancel{+1360.00}\, j - 1280000j + 2560.000}$$

$$\frac{800 + j1600}{3.200.000} = 0.00025 + 0.00005j$$

◆ Igualamos ◆

$$\frac{1}{4000} + j500C = 0.0025 + 0.0005j$$

Admitancia

entonces

$$\frac{1}{4000} = 0.0025$$

$$0.00025 = 0.0025$$

Parte imaginaria

$$500\,C = 0.0005$$

$$C = \frac{0.0005}{5000}$$

$$C = 0.0000001\,F \approx 0.1\,\mu F$$